Numerical Methods - I

Most of the problems in engineering and applied sciences can be ~~formed~~ formulated in terms of system of linear or non-linear equations, ordinary or partial differential equations, integral equations, etc. The majority of these problems do not possess exact or analytical solutions, sometimes solution may be very difficult in all such cases numerical methods provide best approximate solutions.

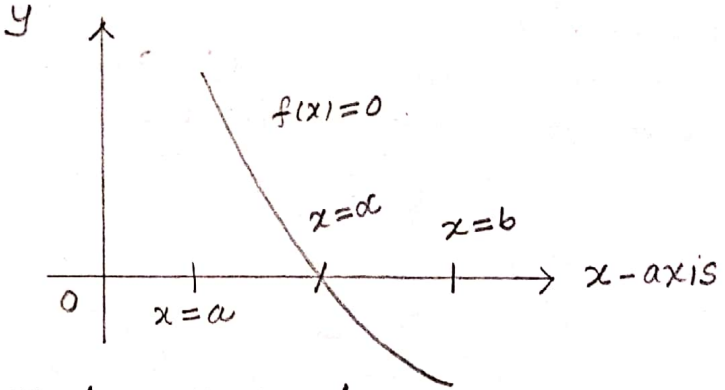
Roots of algebraic and transcendental equations :

Consider a continuous function $y = f(x)$ over (a, b), the equation $f(x) = 0$ is called as algebraic if $f(x)$ is a polynomial, if $f(x)$ involves functions like exponential, logarithmic, trigonometric, hyperbolic, etc equation is called as non-algebraic (or transcendental).

A point $x = \alpha$ is a root (simple root) of $f(x)$ if $f(\alpha) = 0$, geometrically it is a point where curve crosses x-axis (in case of simple root) or touches x-axis (case of multiple roots)

Intermediate value property :

If $y = f(x)$ is a continuous function over $[a, b]$ such that $f(a) \cdot f(b) < 0$, then $f(x) = 0$ has atleast one root (or odd number of roots) in (a, b).



There are several methods to determine roots of equation $f(x) = 0$, each has its advantages and disadvantages. We shall discuss the following methods :

(1) Newton - Raphson method

(2) Regula - Falsi method (or method of false position)

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Newton - Raphson method :

Consider an equation $f(x)=0$ over $[a,b]$ such that $f(a)\cdot f(b)<0$ i.e. $f(a)$ and $f(b)$ should be of opposite sign. Let $x_0 \in (a,b)$ be an approximation for root of $f(x)=0$.

Let $x_1 = x_0 + h$ (1) where h is correction to be determined, such that $f(x_1)=0$

i.e. $f(x_0+h)=0$, by Taylor series we have,

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!}f''(x_0) + \cdots = 0$$

if h is small, $\frac{h^2}{2!}, \frac{h^3}{3!}, \ldots$ are still smaller,

hence, $f(x_0) + hf'(x_0) \approx 0$

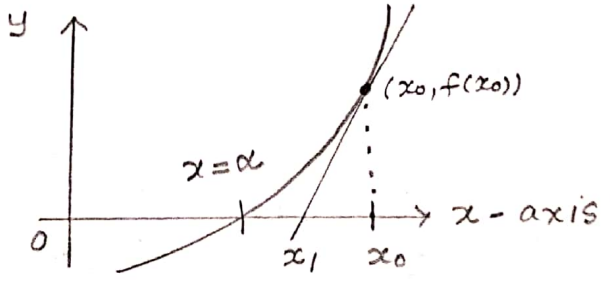
$$\therefore \quad h = -\frac{f(x_0)}{f'(x_0)} \quad (2), \text{ from } (1)$$

$$\therefore \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)},$$

Similarly, $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ and so on

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

The sequence $x_1, x_2, x_3, \ldots$ gives better solution, the process is terminated if two successive values of x_i are same up to four or five places of decimal.



Geometrically, in this method curve $f(x)$ is replaced by tangent at $x=x_0$, x_1 gives improved root and so on till solution $x=\alpha$ can be obtained.

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Examples :

1. Find a positive real root of $x^6 - x^4 - x^3 - 1 = 0$ by using Newton-Raphson method with 1.7364 as initial approximation.

Sol : Here $f(x) = x^6 - x^4 - x^3 - 1$, $x_0 = 1.7364$ (given)

$\therefore f'(x) = 6x^5 - 4x^3 - 3x^2$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.7364 - \frac{f(1.7364)}{f'(1.7364)}$$

$x_1 = 1.54977$,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.54977 - \frac{f(1.54977)}{f'(1.54977)}$$

$x_2 = 1.44313$,

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.44313 - \frac{f(1.44313)}{f'(1.44313)}$$

$x_3 = 1.40734$

Similarly $x_4 = 1.40364$, $x_5 = 1.40362$

Here $x_5 \approx x_4$ $\therefore$ Root of $f(x) = 0$ is 1.40362

In the above example if x_0 is not provided, we need to determine interval containing root, which is done as below:

$f(0) = -1$, $f(1) = -2$, $f(2) = 39$

here $f(1) < 0$ and $f(2) > 0$ $\therefore$ $f(x) = 0$ has root in $[1, 2]$, choose any x_0 such that $1 < x_0 < 2$, the method converges to solution.

The error created at each stage decrease quadratically, hence Newton-Raphson method has quadratic convergence.

2. Using Newton-Raphson method determine square root of a positive number N, hence obtain $\sqrt{35}$ correct to four places of decimal.

Sol : Determining ~~positive~~ square root a number is same as finding real root of second degree equation i.e.

$x = \sqrt{N}$ $\therefore$ $x^2 = N$, $x^2 - N = 0$

$\therefore$ $f(x) = x^2 - N$, $f'(x) = 2x$

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$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= x_0 - \frac{(x_0^2 - N)}{2x_0} = \frac{2x_0^2 - x_0^2 + N}{2x_0}$$

$$= \frac{x_0^2 + N}{2x_0} = \left(x_0 + \frac{N}{x_0}\right)\frac{1}{2}$$

$$\therefore x_1 = \frac{1}{2}\left(x_0 + \frac{N}{x_0}\right)$$

$$x_2 = \frac{1}{2}\left(x_1 + \frac{N}{x_1}\right) \quad \text{and so on}$$

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right) \quad (1)$$

Now to determine $\sqrt{35}$, consider the equation : $x^2 - 35 = 0$, $f(x) = x^2 - 35$

$f(5) = -10 (< 0)$, $f(6) = 1 (> 0)$

$f(x) = 0$ has root in $(5, 6)$

Let $x_0 = 5.5$, using above equation

$x_1 = 5.93182$, $x_2 = 5.91610$

$x_3 = 5.91608 = x_4$

$\therefore \sqrt{35} \approx 5.91608$

3. Find a positive real root of equation $\log(x) = \cos(x)$ correct to four places of decimals.

Let $f(x) = \log(x) - \cos(x)$

$\therefore f'(x) = (1/x) + \sin(x)$

$f(0) = -\infty$ as $\log(0) = -\infty$

$f(1) = -0.54403 (< 0)$, $f(1.5) = 0.33473 (> 0)$

[In case of any trigonometric function 'x' is in radians] Let $x_0 = 1.29341$, $f(x) = 0$ has root in $(1, 1.5)$, you also choose any x_0 in this interval.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.29341 - \frac{f(1.29341)}{f'(1.29341)}$$

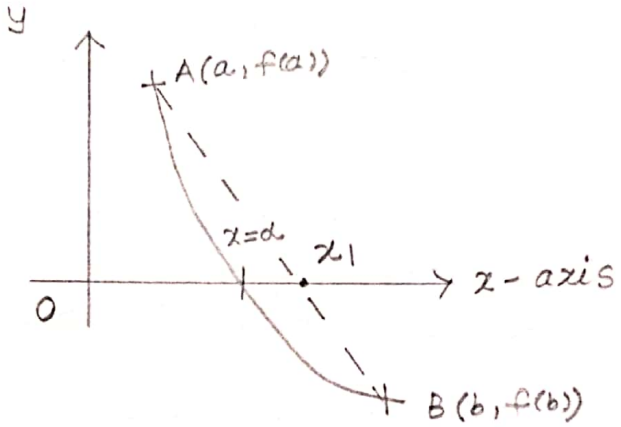
$x_1 = 1.30296$, $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$x_2 = 1.30296$ $\therefore$ Root is 1.30296

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Regula - Falsi method (Method of false position):

The Newton's method of determining root of $f(x)=0$ has a drawback, the point x_0 should be sufficiently close to root $x=\alpha$, method also require large value for $f'(x_0)$, sometimes this method may diverge. Hence, the derivative is replaced by slope of chord joining two points $x=a$ and $x=b$, intercepton of Chord with x-axis is taken as approximation for root, hence name 'method of false position'. This method converges to solution if $x=a$ and $x=b$ are such that $f(a)\cdot f(b)<0$.



Equation of chord AB is:

$$y - f(a) = \frac{f(b)-f(a)}{(b-a)}(x-a)$$

at $x=x_1$ $y=0$

$$\therefore \quad -f(a) = \frac{f(b)-f(a)}{(b-a)}(x_1-a)$$

hence $$-(b-a)f(a) = [f(b)-f(a)](x_1-a)$$

$$x_1 = a - \frac{(b-a)f(a)}{[f(b)-f(a)]}$$

$$\therefore x_1 = \frac{af(b)-bf(a)}{f(b)-f(a)}$$

Now root lies in $[a, x_1]$ or $[x_1, b]$ depending on $f(a)f(x_1)<0$ or $f(x_1)f(b)<0$, Above, algorithm is repeated to get x_2, x_3 and so on, each giving better solution.

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Examples:

1. Find a positive real root of the equation $x \log_{10}(x) = 2$ correct to four places of decimal by Regula-falsi method.

Sol: Let $f(x) = x \log_{10}(x) - 2$

$f(1) = -2 < 0$, $f(2) = -1.39794 < 0$,

$f(3) = -0.56864 < 0$, $f(4) = 0.40824 > 0$

$f(x) = 0$ has root in $[3, 4]$

Let $a = 3, b = 4$, $x_1 = \dfrac{a f(b) - b f(a)}{f(b) - f(a)}$

$\therefore x_1 = \dfrac{3f(4) - 4f(3)}{f(4) - f(3)} = 3.58210$

Now $f(x_1) = f(3.58210) = -0.01502$

the root of equation lies in $[3.58210, 4]$ because $f(3.58210) < 0$ and $f(4) > 0$ not in $[3, 3.58210]$ as $f(3) < 0$ and $f(3.58210) < 0$, ordinates has same sign.

again $x_2 = \dfrac{a f(b) - b f(a)}{f(b) - f(a)} = \dfrac{3.5821 f(4) - 4 f(3.5821)}{f(4) - f(3.5821)}$

$x_2 = 3.59693$, $f(x_2) = -0.00035 < 0$

$f(x) = 0$ has root in $[3.59693, 4]$ as

$f(3.59693) < 0$, $f(4) > 0$

$\therefore x_3 = \dfrac{af(b) - bf(a)}{f(b) - f(a)} = \dfrac{3.59693 f(4) - 4 f(3.59693)}{f(4) - f(3.59693)}$

$x_3 = 3.5973$, $f(x_3) \approx 0$

$\therefore$ Root of given equation is 3.5973

2. Determine a positive real root of $1 - x + \sin(x) = 0$ correct to four places of decimal, using method of false position.

Sol: Let $f(x) = 1 - x + \sin(x)$,

here note that for $\sin(x)$, 'x' is argument in radians.

$f(0) = 1 > 0$, $f(1) = 0.84147 > 0$

$f(2) = -0.09070 < 0$ $f(x) = 0$ has root in $[1, 2]$

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$$\therefore \quad x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{1f(2) - 2f(1)}{f(2) - f(1)} = 1.9027$$

$f(x_1) = f(1.9027) = 0.04272 > 0$

$f(x) = 0$ has root in $[1.90270, 2]$ as $f(x_1) \cdot f(b) < 0$

$$\therefore \quad x_2 = \frac{1.90270\, f(2) - 2f(1.9027)}{f(2) - f(1.9027)} = 1.93386$$

$f(x_2) = f(1.93386) = 0.00095 > 0$

$f(x) = 0$ has root in $[1.93386, 2]$ as $f(x_2) \cdot f(b) < 0$

note that, root does not lie in $[x_1, x_2]$

i.e. $[1.9027, 1.93386]$ as $f(x_1) \cdot f(x_2) > 0$

$$\therefore \quad x_3 = \frac{1.93386\, f(2) - 2f(1.93386)}{f(2) - f(1.93386)} = 1.93455$$

$f(x_3) \approx 0$

$\therefore$ A positive real root of $f(x) = 0$ is 1.93455

Practice problems:

1. Find a positive real root of $xe^x = \cos(x)$ by method of false position correct to four places of decimal. [HINT: Root lies in $[0,1]$, $x = 0.51773$]

2. Using Regula-Falsi method, determine a positive real root of $\cos(x) = 3x - 1$ correct to four places of decimal [HINT: Root lies in $[0,1]$, $x = 0.6071$]

3. Find a positive real root of $x \log_{10}(x) = 1.2$ correct to four places of decimal by Newton-Raphson method.
[HINT: $f(x) = x \log_{10}(x) - 1.2 = 0$ has root in $[2,3]$

$x \log_{10}(x) = 1.2$, using change of base

$x \frac{\ln(x)}{\ln(10)} = 1.2$, $x \ln(x) = 1.2 \ln(10) = 2.7631$

where $\ln(x) = \log_e(x)$; change of base is required

since $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$ and $\frac{d}{dx}[\log_{10}(x)] \neq \frac{1}{x}$

Now $f(x) = x \ln(x) - 2.7631$, $f'(x) = x \cdot \frac{1}{x} + \ln(x)$

solution is 2.7406]

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4. The bacteria concentration C in a reservoir varies as $t^4 - t = C$, calculate the time t required for the bacteria concentration to be $C = 10$, given that the initial time is $t = 2$ s, by Newton's method.

[HINT: $f(t) = t^4 - t - 10$, $t_0 = 2$
$f'(t) = 4t^3 - 1$, Solution is $t = 1.85558$ s]

5. The equation $h^3 - 9h^2 + 3.8197 = 0$ gives the height of a liquid in the spherical tank for the given volume and radius. Using Newton-Raphson method, find the root of the equation near $h_0 = 1$, carry out four approximations.

[$h = 0.67746$]

Methods for Solution of first Order ODE:

The physical process such as motion of objects, heat flow, vibrations, chemical reactions, etc leads to differential equations, analytical solutions (or exact solutions) can be obtained only for selected class of ODE; however numerical solutions can be obtained to desired degree of accuracy with various methods.

We shall discuss the solution of

$$\frac{dy}{dx} = f(x, y), \quad \text{subject to } y(x_0) = y_0$$

by various methods :-

Taylor series method
Modified Euler's method
Runge-Kutta method of order four
Milne's Predictor-Corrector method

Geometrically, the solution of above ODE is a curve passing through the point (x_0, y_0), since it is not possible to integrate above equation in many cases to get the solution, the above methods are used to find best approximations.

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Taylor Series method:

Consider the initial value problem (IVP)

$$\frac{dy}{dx} = f(x,y), \quad y(x_0) = y_0 \qquad (1)$$

Assuming the solution $y = y(x)$ exists of this problem expanding as Taylor series about $x = x_0$, we get:

$$y = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \cdots (2)$$

where $y_0' = \left(\frac{dy}{dx}\right)_{(x_0,y_0)}$, $y_0'' = \left(\frac{d^2y}{dx^2}\right)_{(x_0,y_0)}$, ...

Expression (2) is the Taylor series solution of the problem at a point x in a neighbourhood of the point x_0.

Examples:

1. Use Taylor series method, to find an approximate solution at the point $x = 0.1$, from

$$\frac{dy}{dx} = x - y^2, \quad y(0) = 1$$

Sol: Here $y' = x - y^2$ (1), $x_0 = 0$, $y_0 = 1$

Differentiating w.r.t x successively treating y as function of x, we get:

$$y'' = 1 - 2yy'$$

$$y''' = -2[yy'' + (y')^2]$$

$$y^{IV} = -2[yy''' + y'y'' + 2y'y'']$$

put $x = 0$, $y = 1$ (initial condition), then

$$y_0' = 0 - 1^2 = -1,$$

$$y_0'' = 1 - 2(1)(-1) = 3$$

$$y_0''' = -2[(1)(3) + (-1)^2] = -8$$

$$y_0^{IV} = -2[(1)(-8) + (3)(-1) + 2(-1)(3)] = 34$$

$\therefore$ By Taylor Series

$$y = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \cdots$$

$$y = 1 + (x-0)(-1) + \frac{(x-0)^2}{2!}(3) + \cdots$$

$$y = 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 + \cdots$$

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at $x = 0.1$, $y(0.1) = 0.9138$

2. Use Taylor Series method to find y at the points $x = 0.1$ and $x = 0.2$ given that

$$\frac{dy}{dx} = x^2 + y^2, \quad y(0) = 1$$

Sol: Here $x_0 = 0$, $y_0 = 1$, Taylor series solution of given o.de is

$$y = y_0 + xy_0' + \frac{x^2}{2!}y_0'' + \frac{x^3}{3!}y_0''' + \frac{x^4}{4!}y_0^{iv} + \cdots \quad (1)$$

Since, $y' = x^2 + y^2$,

$$y'' = 2x + 2yy'$$

$$y''' = 2 + 2\{yy'' + y'.y'\}$$

$$= 2 + 2[yy'' + (y')^2]$$

$$y^{iv} = 2[yy''' + y''y' + 2(y')y'']$$

$$= 2[yy''' + 3y'y'']$$

(use of product rule for derivatives)

at $(x_0, y_0) = (0, 1)$, $y_0' = 0 + 1 = 1$

$y_0'' = 2$, $y_0''' = 8$, $y_0^{iv} = 28$,

Using these in (1), we get:

$$y = 1 + x + \frac{x^2}{2}(2) + \frac{x^3}{6}(8) + \frac{x^4}{24}(28) + \cdots$$

$$y \approx 1 + x + x^2 + \frac{4}{3}x^3 + \frac{7}{6}x^4$$

at $x = 0.1$, $y(0.1) = 1.11145$,

$x = 0.2$, $y(0.2) = 1.25253$

Taylor series method for determining solution of IVP involves finding of higher order derivatives could be tedious, hence numerical methods are used to determine solutions, which agrees up to second, fourth order Taylor series are used.

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Modified Euler's method:

Consider the IVP, $\frac{dy}{dx} = f(x,y)$, $y = y_0$ at $x = x_0$

Let $x_1 = x_0 + h$, h is a small step length.

the solution at this point is obtained.

$$y_1 = y(x_1) = y_0 + hf(x_0, y_0) \quad \text{(Euler's method)}$$

This is improved using the modification

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)]$$

this, is further corrected with

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

this process is repeated till two consecutive values of y_1 are same up to desired place of decimal. The above algorithm can be repeated to find solution at point $x_2 = x_1 + h$ and so on.

$\therefore$ If $x_n = x_{n-1} + h$,

$$y_n^{(E)} = y_{n-1} + hf(x_{n-1}, y_{n-1})$$

$$y_n^{(C)} = y_{n-1} + \frac{h}{2}[f(x_{n-1}, y_{n-1}) + f(x_n, y_n^{(E)})]$$

This method has second order accuracy with Taylor series method, sometimes this method is also known as Runge-Kutta method of order 2.

Examples:

1. By using the modified Euler's method, solve

$$y' = \log_e(x+y), \quad y(1) = 2$$

at $x = 1.2$, taking $h = 0.2$, apply correction twice.

Sol: Here: $y' = f(x,y) = \log(x+y)$

$x_0 = 1$, $y_0 = 2$, $h = 0.2$, (given)

$\therefore$ $y_1 = y_0 + hf(x_0, y_0) = 2 + (0.2)\log_e(1+2)$

$y_1 = 2.2197$ (Euler method)

The above solution is improved using modification as follows:

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$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1,y_1)]$$

$$= 2 + \frac{(0.2)}{2}[f(1,2) + f(1.2, 2.2197)]$$

$$= 2 + (0.1)[\log(1+2) + \log(1.2+2.2197)]$$

$$= 2.2328$$

again, $y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1, y_1^{(1)})]$

$$= 2 + \frac{(0.2)}{2}[f(1,2) + f(1.2, 2.2328)]$$

$$= 2 + (0.1)[\log(1+2) + f(1.2+2.2328)]$$

$$= 2.2331$$

Hence, solution at $x = 1.2$ is $y_1 = 2.2331$

2. Using modified Euler's method, solve the initial value problem $\frac{dy}{dx} = \frac{(y-x)}{(y+x)}$, $y(0) = 1$, at $x = 0.1$

Sol: Here $f(x,y) = \frac{(y-x)}{(y+x)}$, $x_0 = 0$, $y_0 = 1$

Let $x_1 = x_0 + h = 0 + 0.1 = 0.1$

Since h is not specified it could be assumed as 0.1 small step size h gives accurate solutions.

$\therefore y_1 = y_0 + hf(x_0,y_0) = 1 + (0.1)f(0,1)$

$$y_1 = 1 + (0.1)\frac{(1-0)}{(1+0)} = 1.1$$

y_1 is improved as below:

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1,y_1)]$$

$$= 1 + \frac{(0.1)}{2}[f(0,1) + f(0.1, 1.1)]$$

$$= 1.09167$$

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + \frac{(0.1)}{2}[f(0,1) + f(0.1, 1.09167)]$$

$$= 1.09161$$

$$y_1^{(3)} = y_0 + \frac{h}{2}[f(x_0,y_0) + f(0.1, 1.09161)]$$

$$= 1.09161$$

Thus, the required solution is

$$y(0.1) \approx 1.09161$$

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Runge - Kutta Method of order four:

The Euler's method and modified Euler method for solving IVP gives poor solutions, since these methods agrees with first order Taylor series solution and second order Taylor series solution respectively. One of the most popular method for determining solution of IVP is Runge - Kutta method of fourth order. It has an accuracy of fourth order Taylor series method without determining higher order derivatives.

Consider $y' = f(x, y)$, $y(x_0) = y_0$

the solution at point $x_n = x_0 + nh$ is constructed as below:

$$y_n = y_{n-1} + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = hf(x_{n-1}, y_{n-1})$

$$k_2 = hf\left(x_{n-1} + \frac{h}{2}, y_{n-1} + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_{n-1} + \frac{h}{2}, y_{n-1} + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_{n-1} + h, y_{n-1} + k_3), \quad n = 0, 1, 2, \ldots$$

Examples:

1. Using the fourth order Runge - Kutta method, solve $(x+y)y' = 1$, at $x = 0.5$, $y(0.4) = 1$

Sol: Here $y' = \frac{1}{(x+y)}$, $x_0 = 0.4$, $y_0 = 1$

$f(x, y) = \frac{1}{(x+y)}$, Let $x_1 = x_0 + h = 0.4 + 0.1$

$x_1 = 0.5$

Compute k_1, k_2, k_3 and k_4 as below:

$$k_1 = hf(x_0, y_0) = (0.1)f(0.4, 1) = 0.07143$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= (0.1)f\left(0.4 + \frac{0.1}{2}, 1 + \frac{0.7143}{2}\right)$$

$$= 0.06731$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$= (0.1)f\left(0.4 + \frac{0.1}{2}, 1 + \frac{0.06731}{2}\right)$$

$$= 0.0674$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.0638$$

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$\therefore$ at $x_1 = 0.5$, $y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$y_1 = 1 + \frac{1}{6}[0.07143 + 2(0.06731) + 2(0.0674) + 0.0638]$$

$$= 0.06744$$

2. Solve $\frac{dy}{dx} = \frac{(y-x)}{(y+x)}$, $y(0) = 1$ at $x = 0.1$, $x = 0.2$ using Runge-Kutta method.

Sol: Here $f(x, y) = \frac{(y-x)}{(y+x)}$, $x_0 = 0$, $y_0 = 1$

Let $x_1 = 0.1$, $x_2 = 0.2$, $h = 0.1$

the solution at $x_1 = x_0 + h = 0.1$ is computed as below:

$$k_1 = hf(x_0, y_0) = (0.1) f(0, 1) = 0.1$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.1) f(0.05, 1.05) = 0.0909$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = (0.1) f(0.05, 1.04545) = 0.09087$$

$$k_4 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_3}{2}\right) = (0.1) f(0.1, 1.09087)$$

$$= 0.08321$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 1 + 0.091125 = 1.091125$$

again at $x_2 = x_1 + h = 0.2$

$$k_1 = hf(x_1, y_1) = (0.1) f(0.1, 1.091125)$$

$$= 0.083209$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = (0.1) f(0.15, 1.13273)$$

$$= 0.07661$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = (0.1) f(0.15, 1.12943)$$

$$= 0.07655$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = (0.1) f(0.2, 1.167675)$$

$\therefore$ $y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$= 1.091125 + 0.07784 = 1.168965$$

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Milne's Predictor - Corrector method:

Euler's method and Runge-Kutta methods are known as single step methods, since the solution at any point uses it previous information i.e. solution at $y_n = y(x_n)$ uses values of $y_{n-1} = y(x_{n-1})$ as input; these methods are generally produce large errors hence multi-step methods are used. In a multi-step method solution at one point say y_4 uses information of solutions at y_0, y_1, y_2 and y_3, one such method is Milne's method. It is a semi-implicit method i.e. solution at one point is predicted and later it improved using corrector. Modified Euler's method is a semi-implict method.

Consider $y' = f(x,y)$, $y(x_0) = y_0$

If the solution at points x_1, x_2, x_3 are available, i.e. $y_1 = y(x_1)$, $y_2 = y(x_2)$ and $y_3 = y(x_3)$ then y_4 is predicted using:

$$y_4^{(p)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

here $f_1 = f(x_1, y_1)$, $f_2 = f(x_2, y_2)$, $f_3 = f(x_3, y_3)$

this value is corrected using

$$y_4^{(c)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(p)})$$

where $f_4^{(p)} = f(x_4, y_4^{(p)})$

above correction is repeated, until the desired order of accuracy is reached.

If y_1, y_2, y_3 are not available, these could be determined using single-step method like Euler's method.

Examples:

1. Use Milne's predictor - corrector method to find y at $x = 0.8$ given $y' = x - y^2$ with $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$, $h = 0.2$

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Sol: Here $f(x,y) = x - y^2$, $h = 0.2$

$x_0 = 0$, $x_1 = 0.2$, $x_2 = 0.4$, $x_3 = 0.6$

$y_0 = 0$, $y_1 = 0.02$, $y_2 = 0.0795$, $y_3 = 0.1762$

We need to find solution at $x_4 = x_3 + h = 0.8$

Now, $f_1 = f(x_1, y_1) = 0.1996$

$f_2 = f(x_2, y_2) = 0.3937$

$f_3 = f(x_3, y_3) = 0.5689$

$$\therefore \quad y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 0 + \frac{4(0.2)}{3}[2(0.1996) - 0.3937 + 2(0.5689)]$$

$$= 0.30488$$

Now $f_4^{(P)} = f(x_4, y_4^{(P)}) = 0.7070$

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 0.0795 + \frac{0.2}{3}(0.3937 + 4 \times 0.5689 + 0.7070)$$

$$= 0.30459$$

Compute $f_4^{(P)} = f(x_4, y_4)$

$$= f(0.8, 0.30459) = 0.70723$$

$$y_4^{(C)} = 0.0795 + \frac{0.2}{3}(0.3937 + 4 \times 0.5689 + 0.70723)$$

$$= 0.3046$$

$\therefore$ Solution at $x_4 = 0.8$ is $y_4 = 0.3046$

2. By employing the Milne's method, solve $y' = 1 + y^2$ at $x = 0.8$, given

$y(0) = 0$, $y(0.2) = 0.203$, $y(0.4) = 0.423$, $y(0.6) = 0.684$

Sol: Here $f(x,y) = 1 + y^2$,

$x_0 = 0$, $x_1 = 0.2$, $x_2 = 0.4$, $x_3 = 0.6$

$y_0 = 0$, $y_1 = 0.203$, $y_2 = 0.423$, $y_3 = 0.684$

y_4 predicted and then improved.

also $x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = h = 0.2$

$f_1 = f(0.2, 0.203) = 1.041$, $f_2 = f(0.4, 0.423) = 1.179$, $f_3 = f(0.6, 0.684) = 1.468$

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$$y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 0 + \frac{4(0.2)}{3}[2(1.041) - 1.179 + 2(1.468)]$$

$$= 1.024$$

Now $f_4^{(P)} = f(0.8, 1.024) = 2.049$

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 0.423 + \frac{0.2}{3}[1.179 + 4(1.468) + 2.049]$$

$$= 1.0297$$

again $f_4 = f(0.8, 1.0297) = 2.0602$

$$\therefore y_4^{(C)} = 0.423 + \frac{0.2}{3}[1.179 + 4(1.468) + 2.0602]$$

$$= 1.0304,$$

compute $f_4 = f(0.8, 1.304) = 2.0617$

$$y_4^{(C)} = 0.423 + \frac{0.2}{3}[1.179 + 4(1.468) + 2.0617]$$

$$y_4^{(C)} = 1.0305$$

Hence the approximate solution at $x = 0.8$ is $y = 1.0305$.

The above ode has exact solution, separating variables:

$$\frac{dy}{dx} = 1 + y^2, \quad \frac{dy}{(1+y^2)} = dx$$

$\therefore$ Integrating, $\tan^{-1}(y) = x + C$

given, $y(0) = 0$ put $x = 0 = y \Rightarrow C = 0$

$\therefore$ $\tan^{-1}(y) = x$ or $y = \tan(x)$ is exact solution

at $x = 0.8$, $y = 1.0296$ (Exact value)

Milne's solution is $y = 1.0305$, (Numerical Solution)

the absolute error is $E_a = |1.0296 - 1.0305|$

$$E_a = 0.0009$$

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Practice problems :

1. Use the Taylor series method, to solve the initial value problem

$$\frac{dy}{dx} = x^2 + y, \quad y(0) = 10$$

at $x = 0.1, 0.2, 0.3$

[HINT: $y \approx 10 + 10x + 5x^2 + 2x^3 + \frac{x^4}{2}$

$y(0.1) = 11.052$, $y(0.2) = 12.2168$, $y(0.3) = 13.5081$]

2. Using the modified Euler's method, find $y(0.1)$, $y(0.2)$ given $\frac{dy}{dx} = 3x + \frac{1}{2}y$, $y(0) = 1$, $h = 0.1$

[$y(0.1) = 1.067$, $y(0.2) = 1.168$]

3. Employ the Runge-Kutta of order four to solve

$$y' = \frac{(y^2 - x^2)}{(y^2 + x^2)}, \quad y(0) = 1$$

at $x = 0.2$, $x = 0.4$, taking $h = 0.2$

[$y(0.2) = 1.19598$, $y(0.4) = 1.3753$]

4. By using Milne's method, solve $y' = 2e^x - y$, at $x = 0.4$ and $x = 0.5$ given $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.04$, $y(0.3) = 2.09$

[$y(0.4) = 2.162$, $y(0.6) = 2.256$]

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